Diego Alejandro Escobar Barahona

Junior Data Engineer | Cloud Data Engineer | Analytics Engineer

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Professional Summary

Junior Data Engineer & Analytics Engineer with enterprise internship experience at **GBM (IBM & Google Cloud Premier Partner)** and experience designing cloud architecture and data contracts for a **Medallion Data Lakehouse on Google Cloud Platform (GCP)**. **Google Cloud Certified — Associate Data Practitioner**, fluent in **English (C1 Advanced, EF SET 66/100)**, with hands-on experience in **BigQuery, Dataform (SQLX), SQL, Python, and Spec-Driven Development (SDD)**, and working knowledge of **Datastream CDC, IBM InfoSphere DataStage, and IBM DB2 LUW**.

Experience

Data Engineer Intern (Digital Solutions Specialist), GBM (IBM & Google Cloud Premier Partner) – San Salvador July 2026 – Oct 2026

- Designed a **client ETL architecture proposal** for near-real-time cloud replication on **Google Cloud** (**BigQuery, Dataform, Datastream CDC, Cloud Storage**), backed by **Lucidchart** dimensional models and **SQL DDL** scripts.
- Developed and executed **ETL integration jobs under technical mentorship** in **IBM InfoSphere DataStage** (Transformers, Lookups, Joins, Merge) across enterprise test datasets and queried **IBM DB2 LUW** on **Linux RHEL**.
- Supported **auditable cloud governance and FinOps budget estimation** across client proposals by co-authoring **GCP security guidelines** (least-privilege **IAM, MFA**, credential rotation, audit logging, data lineage) and sizing cloud infrastructure in **Google Cloud Pricing Calculator**.
- Significantly reduced corporate PDF documentation compilation time and completed assigned engineering tasks in **Azure DevOps** by specifying and validating a **Python CLI automation tool** under **Spec-Driven Development (SDD)** with **Kiro** and supporting team **Git/SSH** workflows.
- **Co-author | OntoTab (GBM):** Executed empirical benchmark experiments under **5×3 nested cross-validation** across **30,000 public credit-risk records** (UCI Taiwan), evaluating ontology-enriched text descriptions encoded via **frozen BGE-M3 (1,024 dims)** across Logistic Regression, XGBoost, and MLP; with XGBoost, text-only ontology representations (R5) reached **0.5069 AP** versus **0.5596 AP** for the engineered numeric baseline (R8) and **0.5568 AP** for the hybrid model (R9), motivating a separate structured numeric channel.

Projects

FERROSTATS — Cloud-Native Medallion Data Lakehouse & Streaming Pipeline on GCP GitHub | Looker Studio BI | Live App

- Architected a cloud-native **Medallion Data Lakehouse (Bronze / Silver / Gold)** in **Google BigQuery** orchestrated via **Google Dataform**, structuring **4 core Bronze tables** (`raw_sales`, `raw_sale_items`, `raw_products`, `raw_inventory_movements`), **3 Silver staging views** (`stg_sales`, `stg_sale_items`, `stg_products`) with automated data quality assertions, and **3 Gold Star Schema analytical marts** (`fct_daily_sales`, `mart_inventory_pareto_abc`, `dim_products`).
- Defined functional data contracts under **Spec-Driven Development (SDD)** for 80/20 Pareto ABC inventory segmentation, Daily Burn Rate, Days of Inventory (DOI), and Reorder Point (ROP), directing and validating AI-assisted SQLX implementations to power a **3-page Google Looker Studio executive dashboard**.
- Specified a decoupled ingestion architecture achieving **<50 ms local ACID checkout in SQLite** with **asynchronous REST streaming into BigQuery Bronze**, establishing transactional concurrency rules and UUID idempotency contracts under **Spec-Driven Development (SDD)**.
- Configured a permanent **$0.00/mo GCP FinOps architecture** (**~$40/mo saved** across **~15,000 req/mo**) via scale-to-zero **Cloud Run** containers (**<3 s** cold starts), keyless IAM Service Account authentication, and automated **Artifact Registry** image pruning (**<0.3 GB**).

Skills

Cloud & Data Warehouses: Google Cloud Platform (BigQuery, Dataform SQLX, Cloud Storage, Cloud Run, Artifact Registry, Looker Studio, IAM, FinOps & Pricing Calculator), working knowledge of Datastream CDC

Data Modeling & Pipelines: Medallion Architecture (Bronze/Silver/Gold), Star Schema (Fact/Dimension Tables), Data Contracts & Quality Assertions, Lucidchart, familiar with IBM InfoSphere DataStage

Languages & Methodologies: SQL (GoogleSQL, CTEs, DDL/DML), Python 3.11 (Pandas, NumPy), Spec-Driven Development (SDD), AI-Assisted Workflows, Bash, PowerShell

Databases & Infrastructure: SQLite, familiar with IBM DB2 LUW and Linux RHEL, Docker, Git/SSH, Azure DevOps

Spoken Languages: Spanish (Native), English (C1 Advanced — EF SET 66/100)

Certifications

Google Cloud Certified — Associate Data Practitioner (ID: 23e3584a246f4c149f26fea03cebcbbb) Oct 2026 – Oct 2029

Google Cloud Skill Badges: Derive Insights from Data | Prepare Data for Looker Dashboards and Reports Aug 2026

EF SET English Certificate — C1 Advanced (66/100: Listening C2, Reading C1, Speaking C1) Jan 2026

Code in Place (Python Programming) — Stanford University 2026

Education

University of the People — Pasadena, California, USA Admitted | Term 3, 2027

Bachelor of Science in Computer Science WSCUC-accredited institution | 100% Online

Technical Baccalaureate in Software Development – Instituto Nacional San Luis | *GPA: 9.5/10.0* Expected December 2026

- **Key Curriculum:** Algorithmic Logic, Relational Databases (SQL), Data Structures, Web Architecture & Software Engineering.